

Sai Yashwanth Dasari

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ABOUT

2025 B.Tech AI/ML graduate specializing in Python backend engineering and applied AI: multi-agent LLM pipelines with LangGraph, RAG systems with FAISS/vector search, async and distributed backend infrastructure, and production-style deployment. 8 independently shipped projects, a 5-month internship at Workcohol, and a research internship at the University of Hyderabad. Co-authored a paper presented at IHCI 2023 on GAN-based audio classification. Open source contributor to fastapi-boilerplate (4 merged PRs to a third-party repository). 398 LeetCode problems solved.

EDUCATION

J.B. Institute of Engineering and Technology

Bachelor of Technology in Artificial Intelligence and Machine Learning | CGPA: 7.88/10.0

Hyderabad, Telangana

2021 – 2025

EXPERIENCE

Software Engineer Intern

Workcohol

Dec 2024 – April 2025

Chennai, Tamil Nadu

- Built **15+ REST APIs** in **Django REST Framework** for a survey platform, in a team of 4 developers
- Optimized **MySQL** queries for **40% faster** response times; designed indexed schema for 10+ question types with complex relationships
- Implemented **JWT authentication** and role-based access control across **3 user roles**

Deep Learning Research Intern

University of Hyderabad

Sep 2023 – Jan 2024

Hyderabad, Telangana

- Co-authored a paper presented at **IHCI 2023** on GAN-based Telugu song pattern recognition, collaborating with a research team of 3
- Built **SVM** and **GAN** models in **TensorFlow** for classification tasks, using feature vectors extracted with Librosa/Praat and processed with Pandas/NumPy

PROJECTS

Fuel Route Optimizer | *Django, Django REST Framework, NumPy, PostgreSQL, OpenRouteService API* | [GitHub](#) | [Live Demo](#)

- Built a fuel-stop optimizer routing across **8,100+ US truck stations** loaded into **PostgreSQL**, via a single OpenRouteService API call
- Used vectorized **NumPy haversine** filtering to narrow candidate stations against the route polyline; designed a **dynamic-programming** shortest-path algorithm over reachable stations within a 500-mile tank range, with a stop-penalty term

CognoDB Graph Database Benchmark | *Python, Neo4j Driver, Cypher, Docker* | [GitHub](#)

Aug 2026

- Built an automated benchmark comparing a graph database against **4 alternatives** (Neo4j AuraDB, Memgraph, FalkorDB, self-hosted Neo4j) on traversal, lookups, aggregation, and concurrent throughput
- Measured **p50/p95 latency** across 100+ iterations per query; diagnosed a driver-protocol bug and a network-RTT-dominated latency floor on cloud free tiers

Insurance Claim Document Processing Pipeline | *FastAPI, LangGraph, Gemini API, PyMuPDF* | [GitHub](#) | April 2026

- Built a **multi-agent LangGraph** pipeline classifying insurance claim PDFs into **9 document types** via Gemini vision, routing to specialized extraction agents
- Added exponential backoff retry logic, graceful JSON fallback, and a keyword-based mock testing layer to avoid burning API quota

Mini Web Crawler | *FastAPI, Playwright, aiohttp, PostgreSQL, Docker* | [GitHub](#) | [Docker Hub](#)

May 2026

- Built an **async BFS crawler** with a dual fetch strategy (aiohttp first, Playwright fallback for JS-rendered pages) and robots.txt compliance with per-domain caching

- Designed a **3-table PostgreSQL** schema with background job tracking and duplicate-crawl detection; containerized and published to Docker Hub

GhostDrop — Anonymous File Sharing | *FastAPI, NGINX, Docker Compose, AsyncIO* | [GitHub](#) July 2025

- Built a horizontally scalable file-sharing platform with **NGINX load balancing** across 3 FastAPI instances, supporting 2GB chunked uploads
- Implemented automated self-cleanup via **asyncio** background tasks; one-command deployment via Docker Compose

Finance Data Processing and Access Control Backend | *FastAPI, SQLAlchemy, JWT, pytest* | [GitHub](#) April 2026

- Built a **3-tier RBAC** finance backend (Admin/Analyst/Viewer) across **15+ REST endpoints** with JWT auth, bcrypt hashing, soft delete, and dynamic filtering
- Wrote **33 passing pytest tests** organized by role, with fixtures and an isolated test database ensuring zero production data contamination

Hotel Booking Analytics & RAG Question Answering System | *FastAPI, LangChain, FAISS, Mistral-7B* | [GitHub](#) 2025

- Built a **RAG pipeline** over 119K+ hotel bookings using **FAISS** vector search (sentence-transformers embeddings) and **Mistral-7B** for natural-language Q&A
- Combined vector similarity search with deterministic queries for revenue, cancellation, and lead-time analytics; exposed 4 FastAPI endpoints with background embedding and chart-based reporting

Smart Task Analyzer | *Django, Django REST Framework, pytest, JavaScript* | [GitHub](#) | [Live Demo](#) Nov 2025

- Built a multi-criteria task prioritization system using the **Weighted Sum Model** (MCDM theory) with 4 prioritization strategies
- Implemented dependency management with forward/backward handling and penalty multipliers; **100% pytest coverage** on the REST API

TECHNICAL SKILLS

Backend & APIs: Python, FastAPI, Django, Django REST Framework, Flask, RESTful API Design, Pydantic, Microservices

AI, ML & LLMs: LangChain, LangGraph, RAG Pipelines, FAISS Vector Search, Embeddings, Prompt Engineering, LLM Integration (Gemini, Claude, Mistral-7B), TensorFlow

Databases: PostgreSQL, MySQL, Neo4j (Cypher), SQLAlchemy ORM, Django ORM, Query Optimization, Schema Design, Indexing

Async & Concurrency: AsyncIO, aiohttp, Background Jobs, Concurrent Request Handling, Event-Driven Architecture

Testing & Security: pytest (90%+ coverage), Unit/Integration Testing, JWT, OAuth2, RBAC, API Security

DevOps & Infrastructure: Docker, Docker Compose, NGINX Load Balancing, Git/GitHub, Linux, CI/CD

Data Structures & Algorithms: Dynamic Programming, Optimization, Graph Traversal, Caching — 398 LeetCode problems solved

Open Source: fastapi-boilerplate contributor (third-party repository) — 4 merged PRs